

The Null Zoo: Size and Power of Backtest-Overfitting Corrections on Synthetic Searches with Known Ground Truth

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September 2026

Abstract

Corrections for backtest overfitting—the deflated Sharpe ratio, the multiple-testing haircut, family-wise adjustments and bootstrap tests—are derived under assumptions that real trading research violates. We introduce the Null Zoo, an open and versioned simulation benchmark that draws complete research searches (20 strategies of 504 daily returns) from eight return families with known ground truth and scores seven validators by size, raw power and size-adjusted power. A lag-one autocorrelation of 0.2 raises the false-positive rate of every test that ignores it, except the already near-zero deflated Sharpe ratio, to between 18.79% and 19.65% at a nominal 5%; deflating the t -statistic by Lo’s autocorrelation-adjusted annualisation factor brings it to 5.52% in this favourable AR(1) case, at a cost of a slight excess size elsewhere. Under negative skew the Student- t tests reject 7.60%; a non-normal standard error and an i.i.d. bootstrap each repair one shape and break another, and no validator is calibrated across skew and fat tails; under fat tails, selection favours series with a large positive outlier. Under i.i.d. normal returns the deflated Sharpe ratio with its authors’ 0.95 acceptance rule fails to accept 90.5% of searches containing a strategy with a true Sharpe ratio of 2, because the rule tests a harder hypothesis than no skill and the skilled strategy raises its own benchmark; size-adjusted, it recovers most of that power (46.9% against 52.6%) but trails the Šidák test by 5.2–18.0 points across families. A bootstrap’s number of resamples materially affects its power. The published results reproduce byte for byte from the released code, and a robustness run with a different generator, 10,000 replications per cell and a finer bootstrap shows the findings are not artefacts of the generator or of Monte Carlo error.

1 Introduction

A researcher who tries many variants of a trading strategy and reports the best one reports a biased number. The best of N strategies with no skill has an in-sample Sharpe ratio that grows with N , roughly in proportion to $\sqrt{2 \ln N}$ times the spread of the individual estimates [4], and the same selection effect pervades published factor research [14]. Several corrections are now standard: the deflated Sharpe ratio [3], built on the probabilistic Sharpe ratio [2]; the multiple-testing haircut of Harvey and Liu [13]; family-wise procedures such as those of Holm [15] and Šidák [27]; and resampling tests in the tradition of White [28], Hansen [12] and Romano and Wolf [25].

Each correction rests on assumptions—normal or independent returns, independent trials, a known number of trials—that real searches violate in predictable ways. Returns are fat-tailed, skewed, volatility-clustered and sometimes autocorrelated; strategy variants share most of their positions and so are correlated. Inference for a single Sharpe ratio under such violations is well

studied [17, 18, 23]. How much each violation distorts each *selection* correction, applied to the best of a search, is rarely answered in a form a practitioner can look up.

This paper answers it with a simulation benchmark, the *Null Zoo*. The benchmark draws complete research searches—20 candidate strategies of 504 daily returns each—from eight return families whose ground truth is known by construction, applies seven validators as their implementations are meant to be called, and records how often each one declares the best strategy skilled. When no strategy has skill, that frequency is the validator’s *size*, which should equal its nominal level. When one strategy has a true annualized Sharpe ratio of 2, it is the validator’s *power*, which we report both raw and size-adjusted. We follow the ADEMP structure of Morris et al. [21] for planning and reporting the study; every rate’s Monte Carlo standard error follows from the stated number of replications.

Four findings stand out.

1. Positive autocorrelation breaks the tests that ignore it. With a lag-one autocorrelation of 0.2, the Šidák-adjusted t test, the haircut, the non-normal-error test and an i.i.d. bootstrap all rejected between 18.79% and 19.65% of skill-less searches at a nominal 5%. Deflating the t -statistic by Lo’s autocorrelation-adjusted annualisation factor brings the size to 5.52%, in the case where the factor’s assumption holds exactly, at a cost of a slight excess size elsewhere.
2. Under negative skew the Student- t tests reject 7.60% at a nominal 5%. The non-normal standard error and the bootstrap each fix one shape and break another. Under fat tails, selection favours series with a large positive outlier (the selected series’ mean skewness is 0.59, against -0.03 for the rest), which the non-normal standard error then mistakes for a property of the returns. No validator is calibrated across skew and fat tails.
3. The deflated Sharpe ratio, with the 0.95 acceptance rule its authors use, rejects at most 1.58% of skill-less searches and, under i.i.d. normal returns, fails to accept 90.5% of searches that contain a strategy with a true Sharpe ratio of 2 over two years. The rule tests a strictly harder hypothesis than “the best strategy has no skill”, and the skilled strategy raises its own benchmark. Size-adjusted, it recovers most of the lost power under i.i.d. returns (46.9% against 52.6%) but trails the calibrated test by 5.2–18.0 points across families: the threshold costs most of the power, the statistic the rest.
4. A bootstrap’s number of resamples materially affects its power. With 400 resamples and a Šidák adjustment for 20 trials, the test can reject only when no resample reaches the observed Sharpe ratio. On the same searches, raising the resamples to 1,999 raises its power from 46.7% to 50.9% under i.i.d. normal returns, with almost no change in size.

We also note an analytic point that affects how such comparisons are read: the haircut of Harvey and Liu [13] is defined with two-sided p -values, and read one-sided it is almost exactly the Šidák-adjusted t test, so comparisons at unmatched sides measure the choice of sides.

The benchmark scores a statistic introduced by its own author, the *luck-equivalent trials* count. As a test it is the Šidák-adjusted Student t test; it reports the same evidence in units of trials. Bartoš et al. [6] argue that simulation studies designed by a method’s developers carry a conflict of interest. We report that conflict, publish every generator, validator and seed, and note that the benchmark’s clearest failure—autocorrelation—applies to the author’s statistic as much as to the others. The published run reproduces byte for byte from its code, and a robustness run with a different generator, five times as many replications and a finer bootstrap shows that the findings are not artefacts of the generator or of Monte Carlo error.

Related work. Arian et al. [1] compare out-of-sample testing schemes—walk-forward, k -fold, purged and combinatorial purged cross-validation—in a synthetic market with known structure and score them by the probability of backtest overfitting [5] and the deflated Sharpe ratio. Their object is the resampling scheme used to produce a backtest; ours is the statistical correction applied to its result. Santoni et al. [26] combine several of the same corrections into one robustness grade and report its discrimination on synthetic markets, including a baseline for the deflated Sharpe ratio alone; they do not report size at a nominal level or power for the individual corrections, which is what a user choosing among them needs. The size and power of multiple-testing procedures under dependence are well studied in statistics; our contribution is a small, open, versioned benchmark that uses the return shapes and search structures of trading research and calls each correction the way it is called in practice.

2 Design

2.1 Aims

The study estimates, for each validator and each return family, (i) the probability that the validator calls the best of N skill-less strategies skilled (size), and (ii) the probability that it rejects when one of the N strategies has real skill (power). A validator is *calibrated* in a family when its size is within Monte Carlo error of its nominal level.

2.2 Data-generating mechanisms

One replication draws a *search*: $N = 20$ return series of $T = 504$ periods, read as two years of daily returns ($q = 252$ periods per year). Every series has population mean zero and population variance one, so every strategy’s true Sharpe ratio is zero. In the power arm, series 0 receives a constant drift of $2/\sqrt{252}$ per period, giving it a true annualized Sharpe ratio of 2; the other series are unchanged. The eight families are:

IID normal. Independent standard normal returns; the case every correction is derived for.

Student t_4 . Student t with 4 degrees of freedom, scaled to unit variance: symmetric fat tails. Its fourth moment is infinite, so sample kurtosis does not converge.

Skew ± 1.32 . A centred, unit-variance lognormal with log-standard deviation 0.4, whose skewness is $+1.32$, and its mirror image (-1.32).

GARCH(1,1). $h_t = 0.05 + 0.10 \varepsilon_{t-1}^2 + 0.85 h_{t-1}$, $\varepsilon_t = \sqrt{h_t} z_t$ [8], with unconditional variance one; each series starts at $h = 1$, $\varepsilon = 0$, without burn-in.

AR(1). $x_t = 0.2 x_{t-1} + \sqrt{1 - 0.2^2} z_t$, started from its stationary distribution.

Two volatility regimes. Normal returns whose standard deviation is 0.50 or 1.32 (0.6 and 1.6 rescaled to unit variance), switching with probability 0.02 each period.

Correlated trials. $x_{k,t} = \sqrt{1/2} f_t + \sqrt{1/2} z_{k,t}$ with a factor f_t common to all strategies, so every pair of strategies has correlation 0.5 and the search contains fewer than 20 effectively independent trials.

Normal variates come from the Box–Muller transform [9]. The published run uses a 32-bit xorshift generator [19]; the robustness run in Section 3.9 uses xoshiro128** [7]. Instrument tests run before any validator is scored. On samples of 50,000 observations, every family’s mean is within 0.03 of zero and its variance within 0.12 of one; the skewed families have skewness beyond ± 1 , the t_4 family kurtosis above 5, the AR(1) family a lag-one autocorrelation within 0.02 of 0.2, the GARCH and regime families positive autocorrelation in squared returns, and the correlated family a pairwise correlation within 0.03 of 0.5. In three representative families (i.i.d. normal, negative skew and correlated trials) the drift gives series 0, and only series 0, an annualized Sharpe ratio within 0.2 of 2. Every cell reproduces from its seed.

2.3 Estimands

In each replication the validator sees all 20 series, selects the one with the highest sample Sharpe ratio, and returns a p -value for the hypothesis that this best strategy has no skill. The estimands are the rejection rates $\Pr(p \leq 0.05)$ under the null arm (size) and the skill arm (power). Because raw power rewards a test for being liberal, we also report size-adjusted power: the power at the cutoff at which the same validator rejects at most 5% of the null arm in the same family. Finally, for the skill arm, we report the rate at which the validator rejects *and* the selected strategy is the truly skilled one, since a rejection that crowns the wrong strategy is a discovery only in name.

2.4 Methods

Let $\widehat{SR}$ be the best strategy’s per-period Sharpe ratio (sample standard deviation with divisor $T-1$), $\widehat{SR}_a = \widehat{SR}\sqrt{q}$ its annualized value, and $\hat{\gamma}_3, \hat{\gamma}_4$ its sample skewness and (non-excess) kurtosis. The validators are:

LET. The luck-equivalent-trials test. Under the null and normal returns, $t = \widehat{SR}\sqrt{T}$ is exactly Student t with $T-1$ degrees of freedom, so one skill-less strategy reaches the observed value with probability $p_1 = \Pr(t_{T-1} \geq t)$. The best of N independent ones does so with probability $1 - (1 - p_1)^N$ [27], the reported p -value. As a test it is therefore the Šidák-adjusted Student t test; the statistic reports the same evidence to users in units of trials, $N_q = \ln(1-q)/\ln(1-p_1)$.

LET-Lo. LET after deflating $\widehat{SR}_a$ by the autocorrelation-adjusted annualisation factor of Lo [18], $[1 + \frac{2\hat{\rho}}{1-\hat{\rho}}(1 - \frac{1-\hat{\rho}^q}{q(1-\hat{\rho})})]^{-1/2}$, evaluated at the best series’ lag-one autocorrelation $\hat{\rho}$ (clamped to $[-0.95, 0.95]$), as the haircut code of Harvey and Liu [13] does. The factor assumes geometric decay of the autocorrelations, $\rho_k = \rho^k$, which the AR(1) family satisfies exactly. With $q = 252$ it is close to $\sqrt{(1-\hat{\rho})/(1+\hat{\rho})}$, the AR(1) long-run-variance correction.

LET-NN. LET with the normal approximation and the non-normal standard error of Mertens [20] evaluated at the estimate: $z = \widehat{SR}\sqrt{T-1} / \sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}$.

HC, HC-Lo. The haircut Sharpe ratio of Harvey and Liu [13]: the two-sided Student t p -value of the best strategy, adjusted by Bonferroni for 20 tests, without and with the Lo factor above. The Holm variant differs only when a strategy with a large *negative* Sharpe ratio has the smallest two-sided p -value; in the robustness run it changed the decision in at most 15 of 10,000 searches in any cell, so we do not tabulate it.

DSR. The deflated Sharpe ratio [3], the probabilistic Sharpe ratio [2] evaluated against a benchmark equal to the expected maximum Sharpe ratio of N skill-less trials, computed with the

Table 1: Size: percentage of skill-less searches (20 strategies, 504 daily returns) whose best strategy is called skilled at the nominal 5% level. Published v0 run, 2,000 searches per cell; HC two-sided as Harvey and Liu [13] define it; bootstrap with 400 resamples.

Family	LET	LET-Lo	LET-NN	HC	HC-Lo	DSR	Boot
IID normal	5.9	6.1	5.9	<i>2.8</i>	<i>3.6</i>	<i>0.0</i>	4.8
Student t_4	5.3	5.5	7.2	<i>2.8</i>	<i>2.8</i>	<i>0.1</i>	6.0
Skew -1.32	8.8	8.8	5.1	5.5	5.5	<i>0.1</i>	5.3
Skew $+1.32$	<i>3.6</i>	3.8	7.0	<i>1.4</i>	<i>1.7</i>	<i>0.1</i>	5.8
GARCH(1,1)	4.6	5.0	4.8	<i>2.4</i>	<i>2.6</i>	<i>0.0</i>	4.8
AR(1), $\phi = 0.2$	20.0	5.9	20.1	13.2	<i>2.7</i>	<i>0.2</i>	17.1
Two vol. regimes	5.1	5.1	5.5	<i>2.5</i>	<i>2.4</i>	<i>0.0</i>	5.1
Correlated, $\rho = 0.5$	<i>3.5</i>	<i>3.5</i>	<i>3.4</i>	<i>1.7</i>	<i>1.9</i>	<i>1.3</i>	<i>3.1</i>

Bold: above the 99% Monte Carlo band around 5% (3.7–6.3); italic: below it. A two-sided $p \leq 0.05$ for a positive Sharpe ratio is a one-sided test at 2.5%, so the HC columns are not comparable with the others here; Table 2 reads them one-sided.

best strategy’s skewness and kurtosis, $N = 20$ and the cross-trial standard deviation of the 20 annualized Sharpe ratios. Bailey and López de Prado [3] accept a strategy when $\text{DSR} \geq 0.95$; we score $1 - \text{DSR}$ as its p -value, so that rejection at 5% is exactly that rule.

Boot. An i.i.d. bootstrap [11] of the best strategy’s demeaned returns. With B resamples, the single-trial p -value is $(h + 1)/(B + 1)$, where h counts resampled Sharpe ratios at or above the observed one, and it is then adjusted by Šidák for 20 trials. The published run uses $B = 400$. The smallest attainable adjusted p -value is then $1 - (400/401)^{20} = 0.0487$, so the test can reject only when no resample reaches the observed Sharpe ratio, which costs power without affecting validity. The robustness run uses $B = 1,999$, whose smallest attainable p -value is 0.0100.

LET, HC, HC-Lo and DSR call the implementations behind the author’s public calculator and API. LET-Lo applies the same implementation to a Lo-deflated Sharpe ratio. LET-NN, Boot and the autocorrelation estimate are written in the benchmark itself. No validator is tuned to the benchmark.

2.5 Performance measures and Monte Carlo error

The published run (v0) uses 2,000 replications per cell and fixed seeds (one per cell, recorded in the output). The Monte Carlo standard error of a rejection rate $\hat{p}$ from n replications is $\sqrt{\hat{p}(1 - \hat{p})/n}$: 0.49 percentage points at 5% for $n = 2,000$ and 0.22 for $n = 10,000$. Every rate’s standard error follows from this formula; the tables state n . We flag a size as miscalibrated when it lies outside the 99% band around 5%, which is 3.7–6.3% in the published run and 4.44–5.56% in the robustness run. With seven validators in eight families, a 99% band keeps the expected number of false flags among the 56 size cells below one.

3 Results

Table 1 reports size in the published run. Tables 2–4 report the robustness run described in Section 3.9. Unless stated otherwise, rates in the text come from it, because its Monte Carlo error

Table 2: Size in the robustness run: xoshiro128** generator, 10,000 searches per cell, HC read one-sided (two-sided p halved), bootstrap with 1,999 resamples.

Family	LET	LET-Lo	LET-NN	HC	HC-Lo	DSR	Boot
IID normal	5.16	5.59	5.03	5.04	5.45	<i>0.02</i>	4.94
Student t_4	4.92	5.12	6.76	4.84	5.00	<i>0.10</i>	6.46
Skew -1.32	7.60	8.07	<i>4.20</i>	7.45	7.93	<i>0.07</i>	4.80
Skew $+1.32$	<i>2.67</i>	<i>2.90</i>	5.30	<i>2.62</i>	<i>2.82</i>	<i>0.05</i>	<i>4.19</i>
GARCH(1,1)	<i>4.40</i>	4.96	4.53	<i>4.33</i>	4.80	<i>0.04</i>	<i>4.42</i>
AR(1), $\phi = 0.2$	19.65	5.52	19.54	19.39	5.39	<i>0.30</i>	18.79
Two vol. regimes	5.25	5.26	5.82	5.15	5.13	<i>0.04</i>	5.35
Correlated, $\rho = 0.5$	<i>3.17</i>	<i>3.27</i>	<i>3.13</i>	<i>3.09</i>	<i>3.22</i>	<i>1.58</i>	<i>3.00</i>

99% Monte Carlo band around 5%: 4.44–5.56. The standard error of any rate $\hat{p}$ is $\sqrt{\hat{p}(1-\hat{p})/10,000}$, at most 0.50 percentage points.

Table 3: Power: percentage of searches in which one strategy has a true annualized Sharpe ratio of 2 and the validator rejects. Robustness run, as in Table 2.

Family	LET	LET-Lo	LET-NN	HC	HC-Lo	DSR	Boot
IID normal	53.0	53.2	53.0	52.7	52.9	9.5	51.8
Student t_4	54.3	54.3	54.1	53.8	54.0	10.9	54.2
Skew -1.32	54.6	54.9	44.2	54.3	54.7	7.9	46.1
Skew $+1.32$	52.8	53.0	62.2	52.4	52.7	11.5	58.3
GARCH(1,1)	55.7	55.2	55.5	55.2	54.9	10.8	53.8
AR(1), $\phi = 0.2$	59.8	34.9	59.7	59.5	34.5	8.2	57.9
Two vol. regimes	53.7	53.9	53.5	53.4	53.7	11.8	52.9
Correlated, $\rho = 0.5$	51.5	51.5	51.4	51.2	51.1	25.1	49.5

Raw power rewards a liberal test; compare columns with Table 4. Monte Carlo standard errors are at most 0.50 percentage points.

is smaller (0.22 percentage points at 5%, against 0.49 in the published run) and its bootstrap has finer resolution.

3.1 The baseline is calibrated

Under independent normal returns the luck-equivalent-trials test is exact by construction: the t -statistic is Student t and the trials are independent. Its size is 5.16% in the robustness run; the published run’s 5.9% lies inside that run’s 99% band (3.7–6.3%) and is consistent with sampling noise. The non-normal-error version (5.03%), the one-sided haircut (5.04%) and the bootstrap (4.94%) are also calibrated here. The Lo-corrected version is slightly liberal (5.59%, just above the band), for a reason discussed below, and the deflated Sharpe ratio rejects almost nothing (0.02%).

3.2 Autocorrelation breaks the tests that ignore it

With a lag-one autocorrelation of 0.2, the tests that treat returns as independent rejected about four times as often as their level: 19.65% for the luck-equivalent-trials test, 19.39% for the one-sided haircut, 19.54% with the non-normal standard error and 18.79% for the i.i.d. bootstrap. The deflated Sharpe ratio, which also ignores autocorrelation, stays far below its level (0.30%) because it rejects almost nothing anywhere. Positive autocorrelation makes the sample Sharpe ratio more variable than the i.i.d. formula assumes, so the t -statistic is too large. The non-normal correction

Table 4: Size-adjusted power: power at the cutoff at which each validator rejects at most 5% of the skill-less searches in the same family (its own null arm). Robustness run.

Family	LET	LET-Lo	LET-NN	HC	HC-Lo	DSR	Boot
IID normal	52.6	51.6	52.9	52.6	51.6	46.9	51.8
Student t_4	54.4	54.0	48.8	54.4	54.0	48.3	46.8
Skew -1.32	48.3	47.5	46.9	48.3	47.5	42.4	46.1
Skew $+1.32$	61.4	60.1	61.3	61.4	60.1	52.7	58.3
GARCH(1,1)	57.1	55.3	56.6	57.1	55.3	49.9	53.8
AR(1), $\phi = 0.2$	34.8	33.3	34.8	34.8	33.3	29.5	n/a
Two vol. regimes	52.9	53.4	51.6	52.9	53.4	46.6	49.8
Correlated, $\rho = 0.5$	58.2	57.3	58.1	58.2	57.3	40.2	56.3

n/a: the validator’s smallest attainable p -value already occurs in more than 5% of the null arm, so no cutoff attains the level. Size-adjusted power is not available to a user, who does not know the family; it isolates how well each statistic ranks skilled searches above skill-less ones. The bootstrap’s p -value is discrete, so its attained null rate falls below 5% in some families (IID normal 4.94%; Student t_4 3.94%; Skew -1.32 4.80%; Skew $+1.32$ 4.19%; GARCH(1,1) 4.42%; Two vol. regimes 4.48%; Correlated, $\rho = 0.5$ 4.49%), and its size-adjusted power is then measured at that lower level.

adjusts for the shape of the marginal distribution, which is normal here, and cannot help. The i.i.d. bootstrap resamples single observations and so destroys the very dependence that inflates the Sharpe ratio. Block and stationary bootstraps [16, 24], and the studentized circular block bootstrap that Ledoit and Wolf [17] propose for Sharpe ratios, are the natural resampling remedies; v0 does not include them.

Deflating the t -statistic by the autocorrelation-adjusted annualisation factor of Lo [18] brings the size to 5.52% for the luck-equivalent-trials test and 5.39% for the haircut. That is inside the 99% band but at its upper edge (the 95% band ends at 5.43%). The result should not be over-read, for three reasons.

- The AR(1) family satisfies the factor’s assumption of geometric autocorrelation decay exactly, so it is the correction’s best case. Dependence of other shapes—a moving average, higher-order autoregression, long memory—would call for a heteroskedasticity- and autocorrelation-consistent variance [22] or a block resampling test, which v0 does not score.
- The correction is not free elsewhere. In the seven families without autocorrelation it raised the size by between 0.01 and 0.56 percentage points. Mainly, the estimated factor adds noise to the t -statistic, which fattens its upper tail even when the true autocorrelation is zero; slightly, the factor is convex in the estimate, so an estimate of -0.05 raises the t -statistic by more than one of $+0.05$ lowers it.
- Its power under autocorrelation (34.9%) is well below the uncorrected test’s (59.8%), but the uncorrected test’s power is bought with a fourfold excess of false positives. Size-adjusted, the two are close (33.3% and 34.8%; Table 4).

3.3 The haircut read one-sided is the Šidák test

Harvey and Liu [13] define the haircut with two-sided p -values. For a positive Sharpe ratio, a two-sided $p \leq 0.05$ is a one-sided test at 2.5%, which is why the haircut columns of Table 1 sit near half the level (2.44% under i.i.d. normal returns in the robustness run). Halving the p -value is not how Harvey and Liu define their method; we do it only to compare tests at a common level.

Read that way, the haircut and the luck-equivalent-trials test are, analytically, nearly the same test. Both use the same Student t tail p_1 of the same statistic; Bonferroni rejects when $Np_1 \leq 0.05$, Šidák when $1 - (1 - p_1)^N \leq 0.05$, and the two disagree only when p_1 falls between 0.002500 and 0.002561. In the simulations they disagreed in at most 47 of 10,000 searches in any cell. This is a known equivalence, not an empirical finding; its practical consequence is that comparisons of these corrections at unmatched sides measure the choice of sides. The benchmark’s seven columns therefore contain five distinct decision rules: the Šidák–Bonferroni t test, the same test with the Lo factor, the non-normal-error variant, the deflated Sharpe ratio and the bootstrap.

3.4 Skew and fat tails

The Student- t tests are liberal under negative skew (7.60% for the luck-equivalent-trials test, 7.45% for the one-sided haircut) and conservative under positive skew (2.67%). With $T = 504$, the t -statistic of a skewed series is itself skewed in finite samples, so one of its tails is heavier than Student t and the other lighter.

The non-normal standard error of Mertens [20] moves the error rather than removing it: it is conservative under negative skew (4.20%) and clearly liberal under symmetric fat tails (6.76%). Under positive skew it was outside the band in the published run (7.0%) and in a first 10,000-replication run with the same design (6.19%), but inside it in the robustness run reported here (5.30%); it is liberal there more often than not. It is not designed to correct the null distribution. Under the null of a zero Sharpe ratio its variance term equals one for every i.i.d. distribution; skewness and kurtosis enter only because the term is evaluated at the estimated Sharpe ratio. Selection then makes that estimate misleading. Under t_4 returns the selected best series has a mean sample skewness of 0.59, against -0.03 for the other series (Table 7): a series tends to win because it had a large positive outlier. Plugging that positive skewness into the formula shrinks the winner’s standard error, and the t_4 distribution’s infinite fourth moment makes the kurtosis term unstable as well.

The bootstrap is within the band under negative skew (4.80%), conservative under positive skew (4.19%) and liberal under fat tails (6.46%). The last is plausibly the same selection effect: resampling a winner that owes its rank to an outlier reproduces the outlier’s influence. No validator in the benchmark is calibrated across all three shapes.

Under GARCH volatility clustering and regime switching every test except the DSR stays within one percentage point of 5% (Table 2); the flagged cells are the non-normal-error test under regimes (5.82%, liberal) and, marginally, the Šidák test, the one-sided haircut and the bootstrap under GARCH (4.33–4.42%, conservative; the Šidák test was 4.96% in the first 10,000-replication run). In both families returns remain a martingale difference sequence with finite variance, so the t -test is asymptotically valid, and two years of daily data are enough for it to be nearly so here.

3.5 Correlated trials make the Bonferroni-type tests conservative

When every pair of strategies has correlation 0.5, the search contains fewer than 20 effectively independent trials, and every test that assumes 20 is conservative: 3.17% for the luck-equivalent-trials test, 3.09% for the haircut and 3.00% for the bootstrap. The unused level costs power: 51.5% raw against 58.2% size-adjusted. An estimate of the effective number of trials would recover it; v_0 does not include one.

3.6 The deflated Sharpe ratio at 0.95 tests a harder hypothesis

Bailey and López de Prado [3] accept a strategy when $DSR \geq 0.95$. Scored this way, the deflated Sharpe ratio rejects at most 1.58% of skill-less searches (in the correlated family, and at most 0.30% elsewhere). Under i.i.d. normal returns it fails to accept 90.5% of searches that contain a strategy with a true annualized Sharpe ratio of 2 over two years (between 74.9% and 92.1% across families; power 9.5% under i.i.d. normal returns, against 53.0% for the Šidák test).

Formally, the rule $DSR \geq 0.95$ tests $H_0: SR \leq SR_0$, where SR_0 is the expected maximum of N skill-less trials, a harder hypothesis than $H_0: SR = 0$ for the best of N . Bailey and López de Prado [3] themselves read the DSR as the probability that the true Sharpe ratio exceeds zero; read that way, a value of at least 0.95 should occur in about 5% of skill-less searches, and it occurred in at most 1.58%, so at its 0.95 point it is strongly conservative. The difference in hurdles is large:

- Under independence and normality, the one-sided 5% critical value for the best of $N = 20$ trials is about 2.80 standard errors.
- The DSR’s benchmark is about 1.90 cross-trial standard deviations: the approximation of Bailey et al. [4]; the exact expected maximum of 20 standard normals is 1.87. The rule then asks for a further 1.645 standard errors of confidence on top of that.
- The cross-trial standard deviation is computed from all 20 strategies, including the best. Under the null it averages 0.99 times the i.i.d. standard error, which gives a hurdle of about 3.53 standard errors. Under the alternative the skilled strategy widens the dispersion to 1.17 times the standard error, and the hurdle to about 3.87: a skilled strategy raises its own benchmark.
- The comparison assumes a dispersion near the standard error. When trials are strongly correlated the dispersion shrinks, and below 0.61 times the standard error the DSR’s hurdle falls under the Šidák value; in the correlated family it is 0.70 under the null, which lowers the hurdle to about 2.97 standard errors, close to the Šidák value, and is why the DSR rejects most there.

The ranking the DSR induces is better than its raw power suggests. Size-adjusted, its power under i.i.d. normal returns is 46.9%, against 52.6% for the Šidák test, and across families it trails by 5.2–18.0 points (Table 4). The 0.95 threshold costs most of the power; the statistic, whose benchmark moves with the dispersion of the whole search, costs the rest.

3.7 Resolution matters for the bootstrap

A bootstrap p -value with B resamples is a Monte Carlo estimate, and a test built on it loses power as B falls [10]. Here the loss is sharp because of the Šidák adjustment. With the published run’s 400 resamples the smallest attainable adjusted p -value is 0.0487, just below the level, so the test rejects only when no resample reaches the observed Sharpe ratio; with 1,999 resamples it rejects only when at most 4 do. To measure the effect without run-to-run noise, we scored the same 10,000 searches per cell with both numbers of resamples, drawing the resamples from separate generator streams. Under i.i.d. normal returns the bootstrap’s power rises from 46.7% to 50.9% (the Šidák t test: 52.6%), under t_4 returns from 49.0% to 53.9%, and under negative skew from 40.9% to 46.0%, while its size moves by less than half a percentage point (4.28% to 4.50% under i.i.d. returns). Comparisons of resampling tests with analytical ones should therefore state the number of resamples.

3.8 Rejecting is not the same as finding

Table 6 in the appendix counts only rejections in which the selected best strategy is the truly skilled one. Under i.i.d. normal returns the skilled strategy is the in-sample best in 81.2% of searches, so even a perfect test would crown the wrong strategy in about one search in five. Autocorrelation widens the gap: the skilled strategy is the best in only 65.4% of AR(1) searches.

3.9 Robustness

The published run is deterministic, and running its code reproduces the published result file byte for byte. The robustness run changes only what could make its numbers artefacts:

- the generator, to xoshiro128** seeded by splitmix32 [7], whose first outputs we checked against the reference implementation;
- the number of replications, from 2,000 to 10,000 per cell, with new seeds;
- the bootstrap’s resolution, from 400 to 1,999 resamples;
- the sides of the haircut, which is also read one-sided.

The families, N , T , the effect size and every validator except the bootstrap’s number of resamples are unchanged, so the run rules out generator and Monte Carlo artefacts, not sensitivity to the design. The qualitative findings of the published run hold, with these differences:

- The i.i.d. normal size of the luck-equivalent-trials test moves from 5.9% to 5.16%.
- The bootstrap gains power with more resamples (Section 3.7), and it is liberal under t_4 returns (6.46%; 5.74% with 400 resamples in a first 10,000-replication run), which the published run’s wider band did not detect (6.0%).
- The Lo-corrected test’s slight excess size under i.i.d. returns becomes visible (5.59%).
- Individual cells near the band edges can move by two or three Monte Carlo standard errors between runs; the non-normal-error test under positive skew is the clearest case. Conclusions in this paper rest on patterns across families and runs, not on single cells.

4 Discussion

Implications for practice. Four suggestions follow from the tables, within the limits below. (i) Check the selected strategy’s returns for autocorrelation before any multiple-testing adjustment. When it is present, deflate the t -statistic for it, with Lo’s factor for autoregressive decay or a heteroskedasticity- and autocorrelation-consistent variance [22] otherwise, and expect a slightly liberal test when the true autocorrelation is zero. (ii) Compare corrections at matched sides and at the same number of resamples; otherwise the comparison measures conventions. (iii) Treat $\text{DSR} \geq 0.95$ as the demanding rule it is: it tests whether the best strategy beats the expected best of N skill-less ones, not whether it has any skill, and over two years of daily data it will pass few strategies with a true Sharpe ratio of 2. (iv) When returns are markedly skewed or fat-tailed, do not rely on any single correction in this benchmark; check the chosen test’s size on simulated searches shaped like one’s own returns, or use a studentized block bootstrap [17], which v0 does not score.

Limitations. The benchmark covers one search shape ($N = 20$, $T = 504$) and one alternative (a single strategy with a Sharpe ratio of 2). Size distortions from skew and autocorrelation depend on T and N in ways the benchmark does not map, and real searches often contain many weakly skilled strategies rather than one strongly skilled one. Each family changes one feature at a time; real returns combine several. The only dependence studied is AR(1), which is the Lo factor’s best case. The resampling methods most used for data snooping—the Reality Check [28], the test for superior predictive ability [12] and the stepwise procedure of Romano and Wolf [25]—resample all strategies jointly and preserve their dependence; v0 does not score them, and its one bootstrap resamples only the selected strategy. The DSR is checked only at its 0.95 point, not across its range as a probability. Finally, a benchmark designed by the author of one of the scored methods cannot be neutral in the sense of Bartoš et al. [6]. We publish every generator, validator and seed, and four of the seven validators call the same code a user of the public implementations runs, but independent replication with independently chosen families would be more convincing than anything we can do ourselves.

Next versions. The benchmark is versioned so that it can grow without rewriting its history. Natural additions are families that combine features (skewed and autocorrelated returns) or have non-AR dependence, a grid over N and T , alternatives with many weakly skilled strategies, block and studentized bootstraps, joint resampling tests of the Reality Check family, and an estimator of the effective number of trials for correlated searches.

5 Conclusion

The corrections practitioners use for backtest overfitting keep their promises in the setting they were derived for and fail in predictable ways outside it. Autocorrelation was the most damaging violation tested: it raised the false-positive rate of every test that ignores it, other than the near-zero DSR, about fourfold, a naive bootstrap included, and Lo’s factor repaired it in the AR(1) case at a small cost elsewhere. Skew and fat tails were the most persistent: no validator in the benchmark handled both, in part because selection itself distorts the moments of the chosen series. Several apparent differences between methods come from how they are read rather than from the methods: the haircut’s conservatism comes from its two-sided convention, the deflated Sharpe ratio’s low power from its 0.95 rule, and the bootstrap’s from its number of resamples. An open benchmark with known ground truth makes such statements checkable, and anyone can rerun or extend it from the published code and seeds.

Code and data availability

All code, result files and the source of this paper are at <https://github.com/arhancanli/null-zoo> (release v1.0.0, MIT licence). The benchmark itself, in `src/`, is copied unchanged from <https://github.com/arhancanli/canlicapital> at commit 4f7838f8db35, where it is maintained; its published result file (SHA-256 e8287f19cf25a7e0858ab12d19e2c2154f1068c5c32459123d68ec3a7cd3ce0f) reproduces byte for byte with `npm run reproduce:v0` under Node.js 24. Every number in the text and tables is generated from the result files by `analysis/tables.mjs`, and none is typed by hand. Continuous integration checks the instrument tests, the byte-for-byte reproduction and that the committed tables match the result files.

Competing interests

The author designed the luck-equivalent-trials statistic scored here and maintains open-source implementations of several of the scored methods. No external funding was received.

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A Additional tables

Table 5 gives the power of the published run, for comparison with Table 3. Table 6 counts correct discoveries in the robustness run. Table 7 reports the two diagnostics used in Section 3: the cross-trial dispersion of Sharpe ratios that sets the DSR’s benchmark, and the sample skewness of the selected series.

Table 5: Power in the published v0 run (2,000 searches per cell; HC two-sided; bootstrap with 400 resamples).

Family	LET	LET-Lo	LET-NN	HC	HC-Lo	DSR	Boot
IID normal	53.3	52.9	52.8	43.0	42.9	9.0	47.1
Student t_4	51.5	51.4	51.5	42.0	42.4	10.7	46.9
Skew -1.32	55.6	56.4	45.1	46.1	46.6	7.9	42.4
Skew $+1.32$	54.9	54.5	65.0	45.0	45.3	12.1	54.4
GARCH(1,1)	54.4	54.6	54.4	44.4	44.5	10.4	47.6
AR(1), $\phi = 0.2$	60.3	35.9	60.4	50.8	26.8	7.1	54.4
Two vol. regimes	53.9	53.8	54.0	45.1	45.0	12.2	48.9
Correlated, $\rho = 0.5$	50.1	50.2	49.9	40.3	40.5	22.5	44.2

Table 6: Correct discoveries: percentage of searches in which the validator rejects *and* the selected best strategy is the skilled one. Robustness run.

Family	LET	LET-Lo	LET-NN	HC	HC-Lo	DSR	Boot
IID normal	50.2	50.2	50.2	50.0	50.0	9.5	49.1
Student t_4	51.6	51.5	50.8	51.3	51.2	10.9	50.9
Skew -1.32	50.2	50.2	41.5	49.9	50.1	7.9	42.9
Skew $+1.32$	51.1	51.2	59.2	50.8	50.9	11.5	55.7
GARCH(1,1)	52.5	52.1	52.2	52.1	51.8	10.8	50.7
AR(1), $\phi = 0.2$	47.5	30.9	47.5	47.4	30.6	8.0	46.3
Two vol. regimes	51.0	51.2	50.7	50.7	51.0	11.8	50.1
Correlated, $\rho = 0.5$	51.0	51.1	51.0	50.8	50.7	25.0	49.1

The skilled strategy is the in-sample best in the following share of searches: IID normal 81.2%; Student t_4 81.0%; Skew -1.32 77.1%; Skew $+1.32$ 83.9%; GARCH(1,1) 81.2%; AR(1), $\phi = 0.2$ 65.4%; Two vol. regimes 80.5%; Correlated, $\rho = 0.5$ 96.9%.

Table 7: Diagnostics from the robustness run. SD/SE: mean cross-trial standard deviation of the 20 annualized Sharpe ratios divided by the i.i.d. standard error $\sqrt{q/T}$, in the null and skill arms. Skew: mean sample skewness of the selected best series and of the other series, null arm.

Family	SD/SE, null	SD/SE, skill	Skew of best	Skew of others
IID normal	0.99	1.17	0.00	-0.00
Student t_4	0.99	1.18	0.59	-0.03
Skew -1.32	0.99	1.18	-1.22	-1.29
Skew $+1.32$	0.99	1.17	1.31	1.28
GARCH(1,1)	0.99	1.18	0.04	-0.00
AR(1), $\phi = 0.2$	1.21	1.37	-0.00	-0.00
Two vol. regimes	0.99	1.18	0.14	-0.01
Correlated, $\rho = 0.5$	0.70	0.94	0.00	0.00